Combating Spatial Disorientation in a Dynamic Self-Stabilization Task Using AI Assistants

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Abstract

Spatial disorientation is a leading cause of fatal aircraft accidents. This paper explores the potential of AI to aid pilots in maintaining balance and preventing unrecoverable losses of control in aerospace conditions that lead to spatial disorientation. We hypothesize that by relying on numerical signals, an AI can offer cues and corrective measures that ameliorate spatial disorientation. A multi-axis rotation system (MARS) was used to gather data from human subjects self-balancing in a spaceflight analog condition with a joystick. Simulated pilot models were then trained over this data to exemplify the performance characteristics of humans with different proficiency levels. Various AI assistant models were then paired with these pilot models to offer corrective cues if loss of control is predicted to be a risk. The performance of these assisted pilot models is assessed in co-performance simulations with a virtual inverted pendulum (VIP) programmed with identical physics. We also assess how the utility of assistant models (trained over MARS data) transfers to environments where the paired pilot was trained over data from humans performing the VIP task directly. Results show which types of models are most successful in correcting pilot actions, and where actions do or do not align with human expectations and abilities, suggesting combinations of techniques that might be most successful in real-time with real human subjects. We conclude by discussing implications for human-AI collaboration and trust in real-time situated tasks.

Introduction

For human experts, maintaining spatial awareness and orientation is a critical capacity in domains like piloting, spaceflight, and even driving. Spatial disorientation has been and continues to be a leading cause of fatal aircraft incidents (Braithwaite et al. 1998; Gibb, Ercoline, and Scharff 2011) and occurs when sensory information (e.g. from the visual, somatosensory, and vestibular systems) is erroneous, which can lead directly to unrecoverable crashes, causing immense injury or loss of life (Newman 2007; Gibb, Ercoline, and Scharff 2011). An AI system, meanwhile, can more effectively use numerical signals to maintain its position in space, and could possibly track the pilot and vehicle's positioning in the relevant orientational plane(s), detect if there is a risk of losing control (Daiker et al. 2018; Zgonnikova, Zgonnikov, and Kanemoto 2016; Wang et al. 2022), and even alert the pilot to make corrective maneuvers.

In this paper, we address this question using a documented, realistic simulation of vehicle control in a spaceflight analog condition, where subjects are deprived of gravitational cues and use a joystick to self-balance while seated in a multi-axis rotation system (MARS) device programmed to behave like an inverted pendulum (Panic et al. 2015; Vimal, DiZio, and Lackner 2017, 2019). We took data from trials involving human subjects attempting to keep the MARS balanced and used it to train "pilot" models that exemplified performance characteristics of humans of various proficiency levels. We then trained a variety of "assistant" models that, due to different training data and techniques, *embody* the problem space in ways that may or may not align with how humans do. The simulated pilots and all assistants were placed in a co-performance simulation with a virtual inverted pendulum (VIP) programmed with physics identical to the physical MARS. Assistants attempted to help the pilot models keep the VIP balanced and avert crashes by offering corrective cues when the pilot risked a destabilizing loss of balance. To test transfer to a different environment, we also ran assistants with pilot models trained over data from humans performing an instance of the VIP task directly. We assess how much each type of assistant helped different pilot models improve. We conclude by discussing the implications for human-AI collaboration and trust when performing real-time situated tasks.

Our novel contributions are: 1) a thorough exploration of AI techniques to model the performance characteristics of real humans in dynamic self-balancing while disoriented, a challenging action-learning task with well-controlled parameters; 2) an assessment of different reinforcement learning and deep learning models' abilities to correct mistakes of pilot models and prevent destabilizing loss of control in the performance of this task; 3) an analysis of the properties of assistant models to inform the construction and deployment of successful assistants in co-performance with real humans.

Related Work

Spatial disorientation and balance An inverted pendulum (with center of mass above the pivot point) is a common model of human upright balance in the study of postural dynamics (Riccio, Martin, and Stoffregen 1992). Panic et al. (2015) and Vimal, Lackner, and DiZio (2016) explored the relevance of the MARS balancing task to the perception of gravitational cues during unstable vehicle control. Subjects were strapped into a MARS device programmed with inverted pendulum (IP) dynamics and instructed to stabilize themselves about the direction of balance (DOB) using a joystick. Because the risk of spatial disorientation-related accidents is heightened when visual information is limited (Lyons et al. 2006; Takada et al. 2009), subjects were blindfolded. Typically, humans rely on gravitational cues when balancing, which are detected by the vestibular and somatosensory systems as participants tilt away from the gravitational vertical, however in spaceflight conditions gravitational cues are not reliable. To create a disorienting spaceflight analog condition, Panic et al. (2017) and Vimal, DiZio, and Lackner (2017, 2019) placed participants in the Horizontal Roll Plane, where they were always perpendicular to the gravitational vertical and no longer tilting relative to it. 90% of participants reported spatial disorientation and in data 100% exhibited characteristic positional drifting (Vimal, DiZio, and Lackner 2019). Participants showed minimal learning and frequent "crashes" (reaching pre-programmed $\pm 60°$ boundaries, after which the MARS automatically reset to the DOB).

In AI, tasks like CartPole and IP balancing are well-known simple use cases in reinforcement learning (Barto, Sutton, and Anderson 1983; Anderson 1989; Florian 2007) that serve as demonstration benchmarks for newer algorithms for continuous control such as SAC (Haarnoja et al. 2018) and DDPG (Lillicrap et al. 2015). These algorithms rely on reward signals extracted directly through observation of applied environmental physics and are demonstrated to be proficient at solving non-linear control tasks like IP balancing. However, due to the different nature of input data (environmental physics vs. sensorimotor input), we hypothesize that they learn to perform the task in very different ways from humans due to their different *embodiment* (see below). In addition to deep learning algorithms, in this paper, we explore the application of RL to AI assistance of human-like behavior in disoriented balancing, as well as approaches such as behavior cloning (Ross and Bagnell 2010) and adversarial inverse RL (Fu, Luo, and Levine 2018).[1]

Embodiment *Embodiment* is often thought of in terms of the physical form an agent takes, wherein the actions it is capable of are conditioned upon the articulators it possesses (Pfeifer and Scheier 2001; Di Paolo and Thompson 2014; Pustejovsky and Krishnaswamy 2021). However, the types of information a system is exposed to, such as through different types of sensors, also condition the relations the system develops between itself and its environment (Cariani 1992). Using machine learning to find patterns in data is one way of modeling such relations, but exposure to different types of data may mean that different models learn to perform the same task in different ways. We adopt a definition of embodiment akin to Ziemke (2001)'s *structural coupling*, where a system embodied in an environment takes environmental

[1]Although the RL approaches we use are also parameterized by multilayer neural networks, we use "deep learning" as a shorthand to refer to non-reinforcement learning algorithms, such as sliding window or sequence modeling approaches.

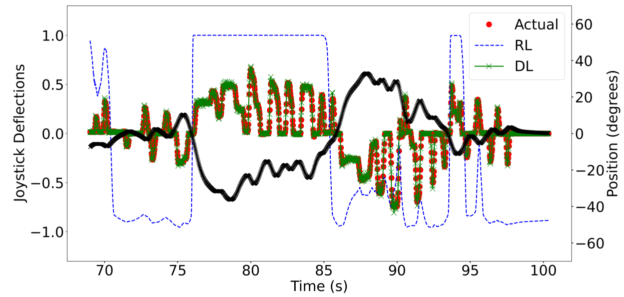

Figure 1: Joystick deflections predicted by a DDPG (blue) and an LSTM trained over human data (green) compared to an actual 30-sec. MARS participant trial sample (participant deflections in red and angular position in black).

states as input, which changes the condition of the system (Quick and Dautenhahn 1999). Differences in how environmental states are measured also result in different outputs or changes to the AI system. Fig. 1 shows a relevant example in our use case: a 30-sec. sample of an actual human trial in the MARS task, with the angular position shown in black and the subject's joystick deflections shown in red, along with the deflections predicted by a DDPG model (blue) and an LSTM trained over actual human motions from other MARS trials (green). While both models learn to perform the task, the model trained on human data *embodies* the problem space similarly to a human, making it a superior predictor of novel humans' actions while solving the same problem. In this paper, we extend prior research on embodiment to assess if and how an AI model's ability to embody a task space similarly to a human is likely to translate into an increased ability to assist a human in the task.

Trust Shneiderman (2020)'s human-centered AI (HCAI) framework separates levels of autonomy and levels of human control in pursuit of systems that achieve high levels of both, such that the human feels in control of task execution at all times, but also achieves better performance with the presence of increased automation than without it. Life-critical situations involving complex systems, such as self-driving cars, military, industrial, and aviation use cases, may require rapid decisions which may have irreversible consequences. Therefore systems deploying high levels of autonomy would need to be *reliable*, *safe*, and *trustworthy* (RST) within known parameters (or limits). Human users must also not become over-reliant on automation such that if the system makes a mistake, the human can still override the AI using human situational knowledge and value judgments. Furthermore, automated systems may prompt over-reliance in emergency scenarios even after the systems have demonstrated a lack of reliability (Robinette et al. 2016; Wagner, Borenstein, and Howard 2018; Tomzcak et al. 2019), or an under-reliance on a typically robust system after a single failure point (Parasuraman and Riley 1997).

Palmer, Selwyn, and Zwillinger (2016) illustrate trust dimensions in the use of an autonomous or automated system such as: *reliability* (low likelihood of failure), *robust-*

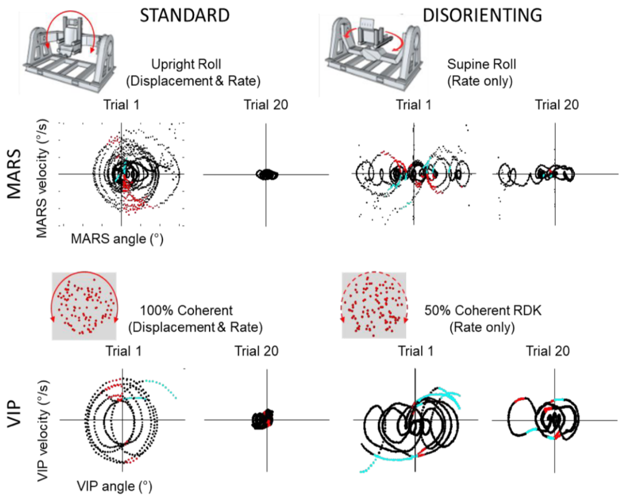

Figure 2: Typical performance in the MARS and VIP tasks, before and after practice. Phase plots show angular velocity vs. angular displacement from the DOB, in 60-sec. MARS and 30-sec. VIP trials. The "standard" conditions provide angular displacement and velocity cues, and subjects improve significantly, seen as clustering around the origin (balance point) by Trial 20. The "disorienting" conditions eliminate sensory signals about displacement from the DOB, increasing positional drift (phase loop oscillations) and destabilizing joystick commands that accelerate away from the DOB in the current direction of motion (cyan dots, where MARS position, velocity and joystick deflection all have the same sign), and minimizing learning.

ness (handling perturbations/deviations appropriately), *understandability* (sensical action), *benevolence* (mission and operator support), and *dynamism* (able to negotiate changes in the environment). Here, we assess the aforementioned dimensions in terms of areas in which our AI assistants are particularly successful or where they fail (reliability), how well they perform when noise is introduced to the simulated interaction (robustness), intuitiveness of actions suggested (understandability), ability to prevent crashes even in circumstances where the unassisted pilot would engage in destabilizing behavior (benevolence), and ability to help pilot models trained over both the MARS and VIP data (dynamism).

MARS and VIP Tasks

The **MARS task** is described above. The data we used was released publicly in Wang et al. (2022), and contains 34 healthy adult subjects (18 female, 16 male, 20.4±2.0 years old). Each subject experienced two experimental sessions on consecutive days, each consisting of 20 100-sec. trials where blindfolded participants attempted to balance themselves with minimal oscillations. The data contains angular positions and velocities, and joystick deflections for each trial at a sampling rate of 50 Hz. MARS dynamics were governed by $\ddot{\theta} = k_P sin\theta$ (θ is degrees deviation from the DOB, pendulum constant $k_P = 600°/s^2$), with an RK4 integrator.

Vimal et al. (2020) used a Bayesian Gaussian Mixture method to cluster subjects into 3 statistically distinct groups that represent Proficient, Somewhat-Proficient and Not-Proficient performance (a.k.a. *Good*, *Medium* and *Bad* proficiency, used hereafter). This was based on features like STD_{MARS} (standard deviation of angular position), number of crashes (excursions beyond ±60°), D_S (short-term diffusion coefficient from the Stabilogram Diffusion Function, or SDF; Collins and De Luca (1993)), $|Mag|_{vel}$ (average magnitude of velocity), *%Destab* (percentage of destabilizing joystick deflections—see Fig. 2), *%Anticipatory* (percentage of anticipatory joystick deflections, where MARS position and joystick deflection have the same sign but the MARS velocity has the opposite sign—usually done to slow the MARS down when velocity is perceived as being too high), Drift Rate, and D_L (SDF long term diffusion coefficient). We follow this performance characterization to split the MARS data into training subsets and use similar features to characterize model performance in the task.

In the **virtual inverted pendulum (VIP)** paradigm, an analog to the MARS written in MATLAB, subjects balance a visually simulated circular array of dots (random dot kinematogram, RDK) which rolls in the plane of the display screen. This is visually rendered for humans and can be directly actuated by an algorithm. In the spaceflight analog condition (similar to the Horizontal Roll Plane in the MARS), the RDK is 50% coherent: alternating subsets of dots displace coherently across consecutive frames while the other half jump randomly, eliminating configural displacement cues relative to the upright DOB while providing low-level retinal motion cues. The physics of the MARS and VIP are identical with the exception of where necessary to keep the weightless VIP consistent with compensations needed for the physical weight of the MARS.

The VIP data consists of 31 healthy adult subjects (22 female, 9 male, 20.2±2.3 years old). These subjects participated in 12 30-sec. trials in a single session in the disorienting condition, with the same objective as the MARS task. The same data was collected at a sample rate of 200 Hz.

We plotted VIP subjects' RMS velocity, velocity SDF diffusion coefficient, and velocity total power vs. fall frequency (# crashes/30 sec.), and found a strong correlation ($r = .730$, $r = .814$, $r = .740$ respectively). Based on a composite of crashes in the 12th trial and crash reduction between Trials 1 and 12 (i.e., final performance and overall improvement), we assigned participants relative rankings and divided them in tertiles to mirror the Good/Medium/Bad MARS classification. VIP participants typically exhibited more crashes, destabilizing actions, and RMS velocity compared to MARS participants of the equivalent proficiency. These factors and differences in sample rate and environment create an interesting testbed for the transfer of our assistants.

Fig. 2 compares the MARS and VIP paradigms. Both can be configured in challenging but non-disorienting modes, with standard sensory information; or difficult and disorienting modes, with degraded information. In both disorienting conditions, subjects show the same characteristic drifting and lack of learning when compared to the coherent con-

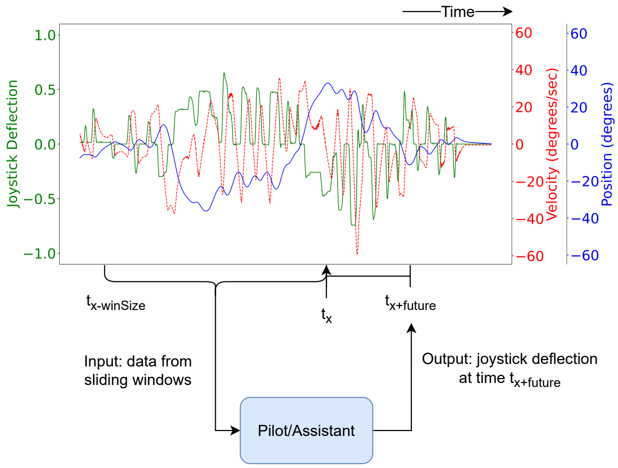

Figure 3: Model input and output structure. "Pilot/Assistant" stands in for any one of the trained prediction models.

ditions.

Model Training

Our goals in model training were both to train models capable of performing an IP balancing task parameterized with MARS physics, and to train models that replicate human performance in the task. In some cases these two categories overlapped, leading to a testable hypothesis: that a model that performs well at the task may also be able to assist a "pilot" (real or simulated) in performing the task better.

Deep Learning Models

To replicate the human-like performance of the MARS task in real-time, we trained Multilayer Perceptron (MLP), Vanilla Recurrent (RNN), Long-Short Term Memory (LSTM; Hochreiter and Schmidhuber (1997)), and Gated Recurrent Unit (GRU; Cho et al. (2014)) network architectures over actions made by humans in the actual MARS data. Input at time t_x is the current angular position and velocity of the MARS and current joystick deflection made by the human, plus a window containing $winSize$ past angular positions, velocities, and joystick deflections. A successful model predicts a joystick deflection made at time $t_{x+future}$. Fig. 3 shows a general schematic. Each type of architecture was trained using different window sizes (0.0s, just the current timestep—MLP models only; and 0.2s, 0.3s, and 0.5s). Architectures with a window size of 0.3s were trained to predict the joystick deflection 0.1s in the future ($future$ value of 0.1s), while others ($future = 0.0$) were trained only to predict the next deflection (so models with $winSize = 0.3, future = 0.1$ directly compare to models with $winSize = 0.2, future = 0.0$). Training data was also split into Good, Medium, and Bad proficiencies and individual models were trained on participants with a specific proficiency. An additional set of models were trained using a combination of 1) Good & Medium and 2) Good, Medium & Bad proficiency data, to see if models could learn strategies employed by certain proficiency groups in scenarios that were not experienced by the other proficiencies. In total, we trained 40 individual DL models. See the appendix for the features of each, and hyperparameter details.

Selecting Pilots All DL models were then made to perform the VIP task (3×30 sec. trials), with angular position, velocity, and joystick deflection recorded at each timestep. We extracted performance features based on those from Vimal et al. (2020), which are shown in Table 1. Since the data distribution could not be assumed to be spherical, these features were used in a k-means clustering process ($k = 3$) to approximate the split into Good, Medium, or Bad groups. Following Vimal, DiZio, and Lackner (2019), the cluster of models that displayed higher oscillations and greater average magnitude of deflections was Bad while the cluster that displayed smaller, more intermittent actions was considered Good (with the remainder considered Medium). We then took the models in each cluster that were trained over the equivalent data subset (e.g., Models in the Good cluster trained over Good data, m.m.), and used the performance characterization technique from the VIP task to identify which model best exemplified the characteristics of each proficiency group: **Good**—LSTM trained over Good data with a window size of 0.2s; **Medium**—GRU trained over Medium data with a window size of 0.3s predicting 0.1s into the future; **Bad**—MLP trained over Bad data with a window size of 0.5s. That each exemplar used a different architecture also suggests that the human subjects exhibited different strategies in performing the MARS task, to different effects. We also trained an instance of the Informer model (Zhou et al. 2021), but this did not perform the task well enough to be a pilot or an assistant and had no bearing on the final results (see appendix).

That these models were trained over MARS data but found to exemplify performance characteristics of different proficiencies when compared to participants in the VIP tasks suggested a level of generalizability between the two different environments. These same architectures were then retrained using data from 3 participants of each proficiency group in the VIP task to produce pilot models trained on VIP data. Table 1 shows the performance characteristics of each pilot exemplar model. The pilot evaluation was performed without any noise added to the actions. All other models that were trained over Good MARS data were reserved to act as candidate assistants while the remaining models were removed from future evaluation.[2]

Reinforcement Learning Models

Models trained over the human data predict what a human would do during MARS/VIP task performance, but this may never lead to an optimal solution to the task itself. Therefore, we also train reinforcement learning-based models that learn directly from exposure to environmental physics using a custom variation of Gymnasium's classic-control Pendulum environment (Towers et al. 2023). The custom environment

[2]The selected Good pilot exemplar architecture was retrained from scratch with different weight initialization to create a distinct instance of the model to act as an assistant.

Pilot	Crashes	% destab.	$\mu\lvert\theta\rvert$ (°)	$\sigma(\theta)$ (°)	$\mu\lvert Mag\rvert_{vel}$ (°/s)	$\sigma(\lvert Mag\rvert_{vel})$ (°/s)	vel RMS	$\mu\lvert d\rvert$
Good	7 / 17	15.9 / 54.0	16.6 / 20.3	21.5 / 23.7	53.3 / 53.0	69.1 / 73.3	70.6 / 77.7	.24 / .11
Med.	9 / 40	21.4 / 63.3	20.1 / 28.8	18.9 / 29.7	68.3 / 122.7	89.3 / 149.4	93.9 / 152.3	.36 / .77
Bad	27 / 23	36.7 / 52.8	21.4 / 19.4	25.9 / 14.3	114.1 / 57.4	121.9 / 72.3	135.9 / 92.2	.50 / .18

Table 1: Performance statistics of pilot exemplar models (values are averaged over 3×30 sec. trials except # crashes, which is summed). Slashes separate models trained over MARS and VIP data. Columns from L–R: # crashes, % destabilizing actions, mean and SD distance from DOB, mean and SD angular velocity magnitude, RMS velocity, and mean deflection magnitude.

includes 1) a problem space analogous to the MARS/VIP task, i.e., bounded at $\pm 60°$ from the DOB; 2) a random starting point for the inverted pendulum within the newly defined problem space; 3) a custom reward function[3] given by $r = \begin{cases} 0, & \text{if } -30° \leq \theta \leq 30° \\ -(\theta^2 + .1\omega^2 + .01d^2), & \text{if } \theta < 30° \cup \theta > 30° \end{cases}$ to encourage small continuous adjustments, such as those of Good MARS participants (Vimal, DiZio, and Lackner 2019; Vimal et al. 2020). The RL algorithms are directly exposed to environmental physics, unlike the DL models which receive only implicit physics through the human performance data. We also train and evaluate behavior cloning (BC) and adversarial inverse RL (AIRL) using Good MARS participant data, to teach the models strategies closer to what humans would execute, in terms of replicating behavior or uncovering implicit reward functions in the data.

RL models take the current angular position and velocity to predict the next joystick deflection ($winSize = 0.0, future = 0.0$, cf. Fig. 3). We use Stable-Baselines3's SAC and DDPG implementations (Raffin et al. 2021), and train them with the default MLP policy, BC, or AIRL. Gaussian distributed noise is added to the action space to encourage exploration as the IP is considered an under-actuated task (Hollenstein et al. 2022). We train: 1) SAC & DDPG with the standard policy; 2) SAC & DDPG with BC; 3) SAC with AIRL. RL hyperparameter details are in the appendix.

Evaluation

Evaluation pairs a pilot model with a candidate assistant model and engages them in co-performance of the VIP task. Fig. 4 illustrates the pipeline with its 4 major components: 1) VIP 2) Pilot; 3) Assistant; 4) Crash predictor.

The **VIP** component is *PyVIP*, a faithful Python port of the VIP task for easier integration and testing of ML models. The **Pilot** model controls the VIP to keep it upright or at least within 60° of the DOB. We used 6 pilot models—each of the architectures selected as performance exemplars, trained over both the MARS and VIP datasets. The **Assistant** model observes the pilot performing the task and will make suggestions when certain conditions are met. In these experiments, the pilot is parameterized with an 80% probability of accepting an assistant suggestion and executing it instead of the pilot's own next action. If accepted, the pilot will perform suggested deflections with the noise of $\mathcal{U}(-.05, .05)$ added

[3]Where θ is the angular position, ω is the angular velocity, and d is the joystick deflection.

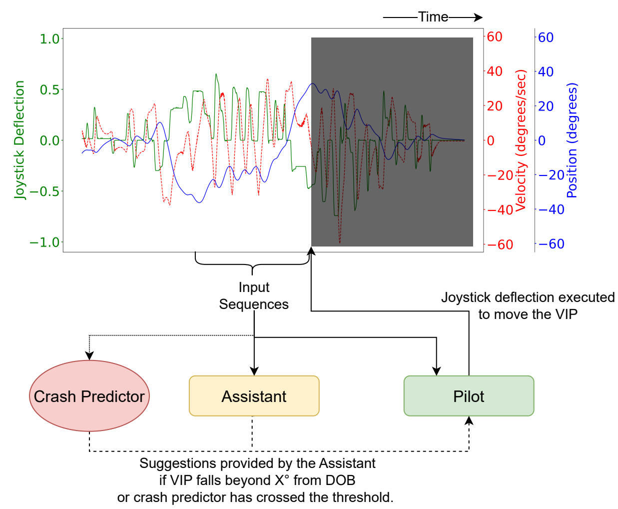

Figure 4: PyVIP evaluation pipeline.

to simulate human imprecision after a $.4+\mathcal{U}(-.05, .05)s$ delay to simulate human reaction time. The **Crash predictor** is a trained instance of the best crash prediction architecture reported in Wang et al. (2022) which is a stacked GRU that takes in similar windows of sequences as the deep learning models and predicts the likelihood of a crash occurring. One caveat of this crash predictor is its high false positive rate. We, therefore, added a *crash probability threshold* of 0.8 where only highly imminent danger would permit assistant suggestions. The assistant would provide suggestions when either 1) crash probability is greater than the threshold *and* angular distance from the DOB exceeds 12°, or 2) angular distance from the DOB exceeds 15°.

We ran each evaluation for 3 30-sec. trials for consistency. Data is sampled at 200 Hz with each sample comprising of the angular position and velocity of the VIP, joystick deflection, crash probability, pilot and assistant's joystick deflections, which entity's deflection was actually performed, and whether the deflection made was destabilizing. 468 individual trials were collected, or 3.9 hours of data.

Results

Table 2 shows performance differences between the pilots when unaided and when aided by different assistants. Overall, SAC-AIRL is the strongest performer, decreasing crashes, % destabilizing deflections, and RMS velocity to a

Assistant	Crashes	% destab.	$\mu\|\theta\|$ (°)	$\sigma(\theta)$ (°)	$\mu\|Mag\|_{vel}$ (°/s)	$\sigma(\|Mag\|_{vel})$ (°/s)	vel **RMS**	$\mu\|d\|$
SAC	0 / -5 / -25 -8 / -25 / -12	2.2 / 4.9 / -18.3 -31.8 / -38.3 / -28.5	0.4 / 1.2 / -3.6 -2.0 / -7.6 -1.0	-0.9 / -6.6 / -7.9 -1.9 / -6.7 / 7.9	18.0 / -25.3 / -56.9 27.0 / -37.7 / 7.5	18.4 / -36.6 / -54.3 27.8 / -48.6 / 11.9	18.7 / -38.7 / -67.5 24.4 / -42.2 / -6.9	0.2 / 0.1 / 0.0 0.4 / -0.1 / 0.3
SAC-AIRL	-2 / -7 / -17 -15 / -33 / -12	1.9 / -3.8 / -14.4 -36.8 / -41.5 / -28.5	1.9 / -0.1 / 5.4 -0.5 / -6.1 / 0.9	0.9 / -7.7 / -0.3 -6.8 / -10.9 / 9.1	-9.8 / -38.7 / -62.8 -22.5 / -66.6 / -8.7	-11.0 / -49.8 / -61.0 -26.1 / -71.7 / -19	-11.2 / -53.5 / -71.2 -29.9 / -72.9 / -20.3	0.1 / -0.1 / -0.0 0.2 / -0.2 / 0.2
DDPG	1 / -2 / -21 -11 / -23 / -12	2.3 / 3.7 / -17.3 -36.4 / -36.4 / -25.1	2.4 / -2.4 / -2.2 2.6 / -7.5 / -1.0	2.1 / -3.1 / -3.4 3.5 / -3.6 / 9.2	25.3 / -11.0 / -43.1 47.4 / -34.1 / 5.5	22.4 / -19.7 / -38.2 47.3 / -36.6 / 7.24	24.6 / -21.2 / -51.0 43.5 / -38.6 / -12.5	0.2 / 0.1 / 0.1 0.6 / -0.1 / 0.3
MLP-GMB-0	-2 / 4 / -11 -1 / -18 / -4	-0.5 / 7.0 / -8.8 -21.6 / -29.7 / -21.5	2.4 / 3.1 / 2.2 2.1 / -6.8 / 4.0	2.2 / 2.8 / -1.7 0.8 / -2.5 / 12.6	18.8 / 7.0 / -43.2 31.4 / -15.5 / 12.2	25.0 / -5.1 / -36.3 41.5 / -20.3 / 22.2	24.5 / 1.3 / -48.0 38.7 / -18.6 / 3.8	0.2 / 0.0 / -0.0 0.3 / -0.1 / 0.2
RNN-G-0.2	5 / 11 / -4 6 / -15 / 0	7.5 / 18.6 / -3.4 -15.7 / -23.1 / -8.2	4.4 / 2.6 / -1.0 0.1 / -7.4 / 0.2	3.4 / 1.8 / -2.1 0.3 / -4.0 / 9.8	24.8 / 13.9 / -24.3 44.8 / -16.2 / 20.4	37.1 / -10.0 / -15.2 53.9 / -12.8 / 35.2	37.6 / 10.8 / -24.6 51.4 / -14.5 / 19.4	0.1 / 0.0 / -0.0 0.3 / -0.1 / 0.2
LSTM-G-0.2	3 / 14 / -7 4 / -13 / 1	8.0 / 23.5 / -1.6 -11.0 / -20.5 / -8.8	3.6 / -0.2 / 0.9 1.7 / -7.3 / 3.2	2.8 / 0.6 / 1.5 2.7 / -4.0 / 12.6	25.4 / 15.6 / -33.7 32.9 / -13.2 / 27.4	33.1 / -15.2 / -22.5 43.8 / -24.4 / 43.6	34.6 / 15.2 / -33.2 41.5 / -14.2 / 24.8	0.2 / -0.0 / -0.0 0.2 / -0.1 / 0.2
GRU-G-0.2	7 / 6 / -8 2 / -14 / 5	5.8 / 8.7 / -7.5 -15.3 / -24.9 / -11.9	3.1 / 4.0 / 0.1 -1.2 / -6.9 / 0.8	3.6 / 2.8 / 0.9 -0.5 / -3.1 / 10.6	25.9 / 5.5 / -32.2 40.8 / -7.0 / 36.4	33.5 / -11.3 / -19.9 62.1 / -18.3 / 42.5	33.5 / -1.0 / -32.5 58.3 / -7.8 / 30.8	0.2 / 0.0 / -0.0 0.3 / -0.1 / 0.2

Table 2: Differences in performance with and without assistance (e.g., *0* means no change in that metric). In each cell, top line refers to MARS pilot models and bottom to VIP pilot models. Slashes separate Good/Medium/Bad pilot models. Under Assistant, G/M/B denotes the proficiency of the assistant training data, decimals denote window size. Shown are the best assistant for ≥1 pilot, or the best assistant *of the given architecture* for ≥3 pilots. See appendix for results for all 26 assistants.

statistically significant level (all $p < 0.0001$ according to a paired two-tailed t-test). A decrease in the metric usually means improvement, e.g., more time spent near the DOB, less oscillation, slower motion, smaller deflections, etc. However, the strongest metric—number of crashes—may be decreased significantly even if some other metrics rise. Performance improvements with strong assistants are greater for less proficient pilot models. By the metrics, RL models are generally better assistants than DL models over human data. The best-performing recurrent models used a window size of 0.2s. Interestingly, MLP-GMB-0 (MLP trained overall proficiencies with no window) decreased crashes as much as SAC-AIRL, but for the Good MARS pilot exemplar only. The Medium exemplar architecture performs significantly worse when trained over VIP data than over MARS data, and the Bad VIP pilot performs much more like the Medium MARS pilot, while Good pilot models are roughly consistent with each other across tasks, suggesting that there are many strategies that lead to poor task performance and relatively few that do well. This also speaks to SAC-AIRL's ability to reduce crashes in both tasks for all proficiency levels. Other high-performing models also reduce crashes (DDPG: $p = 0.0002$, MLP-GMB-0: $p = 0.0033$) and destabilizing deflections (DDPG: $p = 0.0006$, MLP-GMB-0: $p = 0.0080$) for MARS and VIP models, demonstrating transfer between task environments.

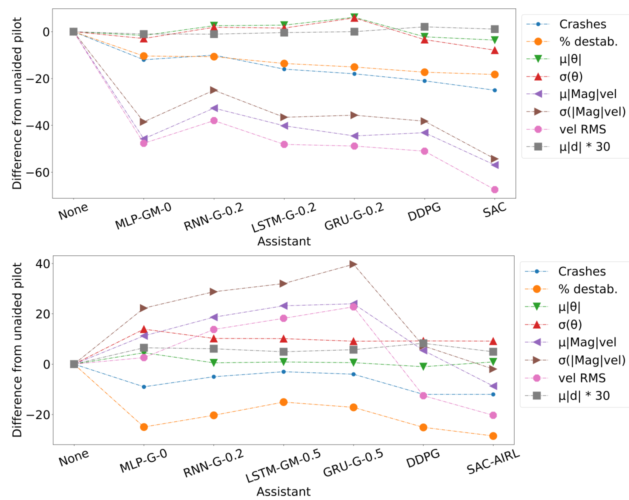

Figure 5: Performance difference for Bad MARS (top) and Bad VIP (bottom) pilot with best assistant of each type. Mean deflection magnitude is multiplied by 30 for visibility and to map the $[-1, 1]$ deflection range to the joystick's physical $\pm 30°$ limits.

Discussion

Fig. 5 shows the difference in performance of both Bad pilot models when assisted by the best assistant of each model type (chosen according to number of crashes). By the metrics, plain SAC is a better assistant for the MARS pilot and SAC-AIRL is better for the VIP pilot. However, in Fig. 6 we see that plain SAC makes suggestions that often rapidly alternate between extremes, whereas SAC-AIRL's suggestions tend to be lower-magnitude, and more intermittent, which is characteristic of the Good human data it saw in training.

Fig. 7 compares SAC-AIRL and MLP-GMB-0, which were competitive in helping the Good MARS pilot model. SAC-AIRL shows its characteristic moderate-magnitude and intermittent cues, while MLP-GMB-0's prediction line closely tracks the actual deflections made, with the greatest divergences still being of low magnitude. This human-data model is a good predictor of human actions and also provides at most small "nudges" based on the multiple proficiencies seen in training, making it a valuable assistant for a pilot who is already good at the task but may need those nudges, while being less helpful for poorer performers.

Conclusion

In this paper, we have explored the space of AI assistance in helping humans maintain balance in a disorienting con-

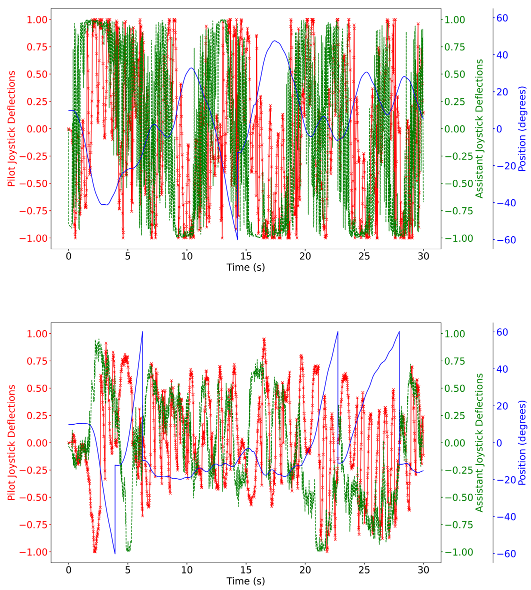

Figure 6: Bad MARS pilot with SAC (top) and Bad VIP pilot with SAC-AIRL (bottom) assistant. Pilot deflections are shown in red and assistant deflections in green.

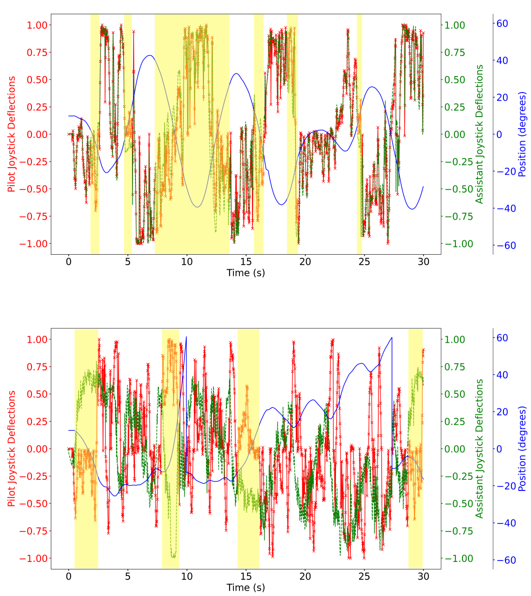

Figure 7: Good MARS pilot with MLP-GMB-0 (all proficiency) (top) and SAC-AIRL assistant (bottom) assistant. Yellow highlights areas where the assistant suggestion diverges from the pilot model's actions.

dition. A combination of RL techniques with strategies to learn from human performance data appears to be most effective in preventing losses of control leading to crashes, due to *embodying* the problem space in a way that incorporates both physics and human signals. We see that although RL models on average perform better as assistants than DL models trained over human data, they do so by suggesting actions that often diverge significantly from the apparent model of the task captured in human actions. This has some significant implications for trust in the model according to the Palmer, Selwyn, and Zwillinger (2016) dimensions. While suggested actions may be appropriately corrective (benevolence), respond to pilot-induced perturbations such as ignoring cues (robustness), and transfer between the MARS and VIP tasks (dynamism), they may not be fully understandable when they diverge significantly from the pilot's internal model of the situation, such as a correction directly opposed to what the pilot would do next.

If AI is already capable of maintaining balance itself, why not have the AI override pilot inputs and take control of a vehicle directly if it detects an imminent loss of control? In a real-life piloting scenario, there may be external factors that still require human value judgments, such as engagement with other vehicles. This necessitates that a human remains in control, but also that the human be assured that the signals received from the AI assistant are verifiable and trustable, to better inform just-in-time decision making.

Future Work

To conduct high-throughput experiments in this space, we have used ML techniques to simulate humanlike behavior, informed by performance characteristics grounded in the domain literature. Therefore the next obvious steps are to evaluate if our findings here can also be applied to novel human subjects who may display traits not captured in the MARS performance data. A drawback of the RL algorithms is that the learned strategy may solve the problem in a highly unhumanlike way, i.e., even though we built approximations of human limitations into the simulation, a human receiving a trained agent's directions might still not be able to follow them correctly. One possible way to address this would be to apply RLHF (Christiano et al. 2017) to this domain through further human-in-the-loop training, allowing the AI and human partner each to improve trial-by-trial, and thereby teach each other progressively better. A future study can also investigate transfer to more complicated conditions, like orientation in multiple roll planes or flight simulators. There also remains the question of *how* to deliver an AI assistant's cues to a human pilot. Possibilities include aurally rendered tones or visual indicators on the screen to indicate the direction and magnitude of the corrective action, or linguistic instructions. Precisely when and how to deliver corrective information is another avenue of future study.

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